REVIEW PDF

COF-System AI4S

Knowledge-driven intelligent assistant for COF design

This reviewer brief condenses the system demonstration video into a compact PDF. It shows how a user moves from COF questions and monomer choices to graph-traced answers, literature evidence, design analysis, and 3D molecular preview.

3
functional modules

508
monomer records

507
3D-ready records

771
COF/dimer candidates

PURPOSE

When the original video cannot be uploaded, the reviewer can still understand what the system does, what evidence is exposed, and how the interface supports COF design.

Knowledge QA

Natural-language COF question answering with graph triples, raw subgraph JSON, generated Cypher, and literature fragments.

COF Design Analysis

Node-linker or monomer-pair analysis with staged checks for reactivity, formability, stability, properties, and HER potential.

Monomer Library

Searchable monomer records with molecular metadata, 3D structure preview, and related synthesizable COF/dimer candidates.

REVIEWER READING PATH

Page 2 explains the dual-evidence architecture. Pages 3-5 show the three major interfaces. Page 6 turns the 516-second video into a concise storyboard.

Evidence Architecture

The system is organized around dual evidence: a literature-RAG path and a knowledge-graph retrieval path.

1. Input

A COF question, or a monomer pair with linkage and reaction conditions.

2. Parsing

The system checks domain scope and extracts monomers, linkages, applications,

3. Retrieval

Neo4j graph relations and local literature fragments are retrieved as evidence.

4. Output

The answer or design recommendation is displayed with traceable evidence panels.

DUAL-EVIDENCE TRACEABILITY

Generated answers are not shown as black-box text. The interface places the final response next to the evidence used to produce it, so reviewers can inspect both the paper-level context and the structured domain relations.

LITERATURE RAG

Paper snippets ground the response.

KNOWLEDGE GRAPH

Triples show source, relation, and target.

VISIBLE AUDIT TRAIL

Inspect raw JSON and generated Cypher.

KNOWLEDGE QA WORKFLOW

Classify the question, parse entities, retrieve graph and literature evidence, then synthesize an answer.

COF DESIGN WORKFLOW

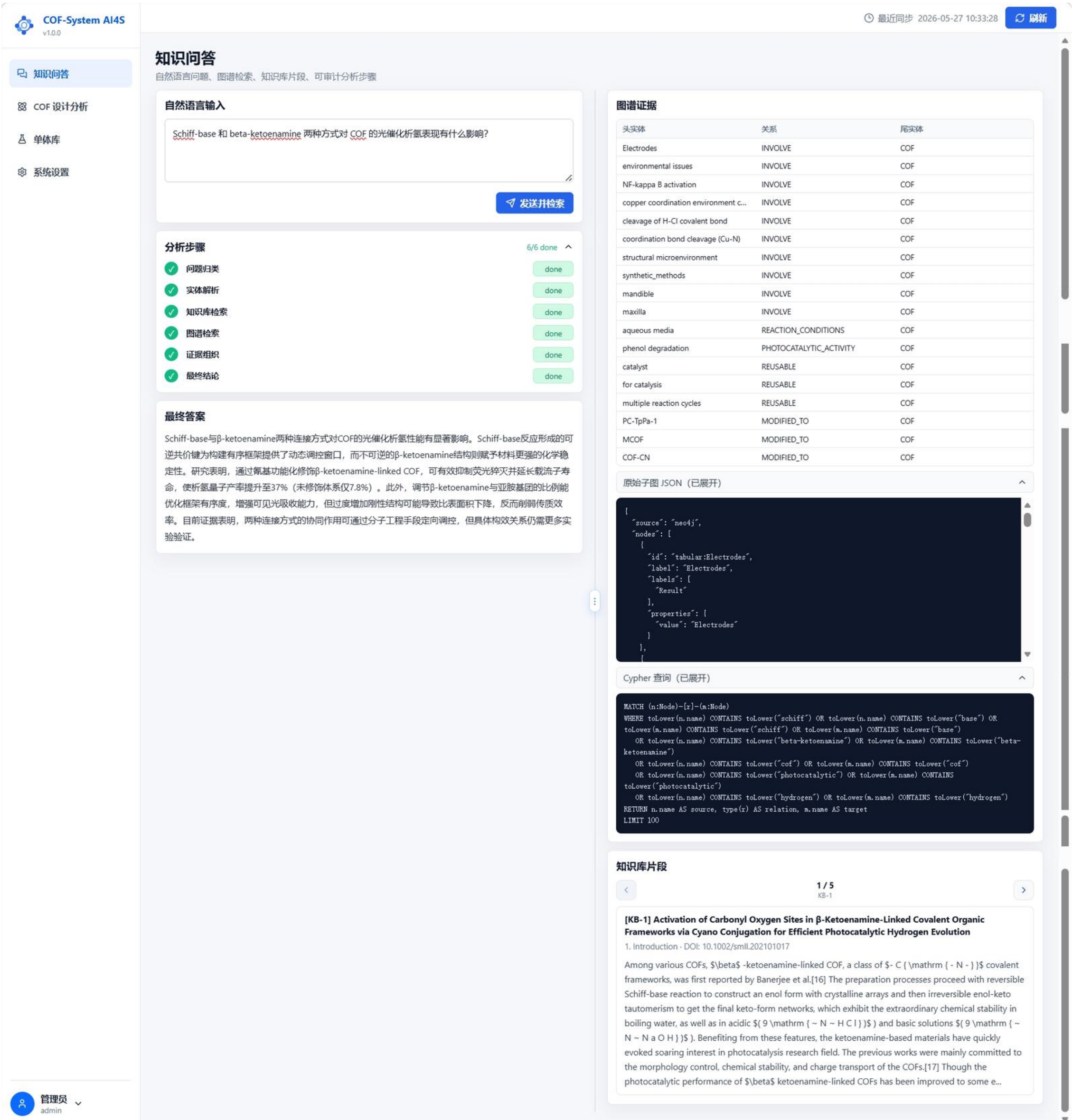
Resolve monomers, evaluate chemistry and properties, then return an evidence-supported design recommendation.

IMPLEMENTATION SIGNALS VISIBLE IN THE DEMO

The web interface includes dedicated pages for QA, COF design analysis, monomer library exploration, and settings. Neo4j provides structured graph relations; the literature retriever provides local paper fragments; the 3D viewer supports monomer and dimer visualization where structures are available.

1. Knowledge QA

A reviewer can ask a COF-related question in natural language and inspect how the answer is grounded in graph and literature evidence.



INPUT

Submit a natural-language COF question.

ANSWER

Read the synthesized response after retrieval.

TRACEABILITY

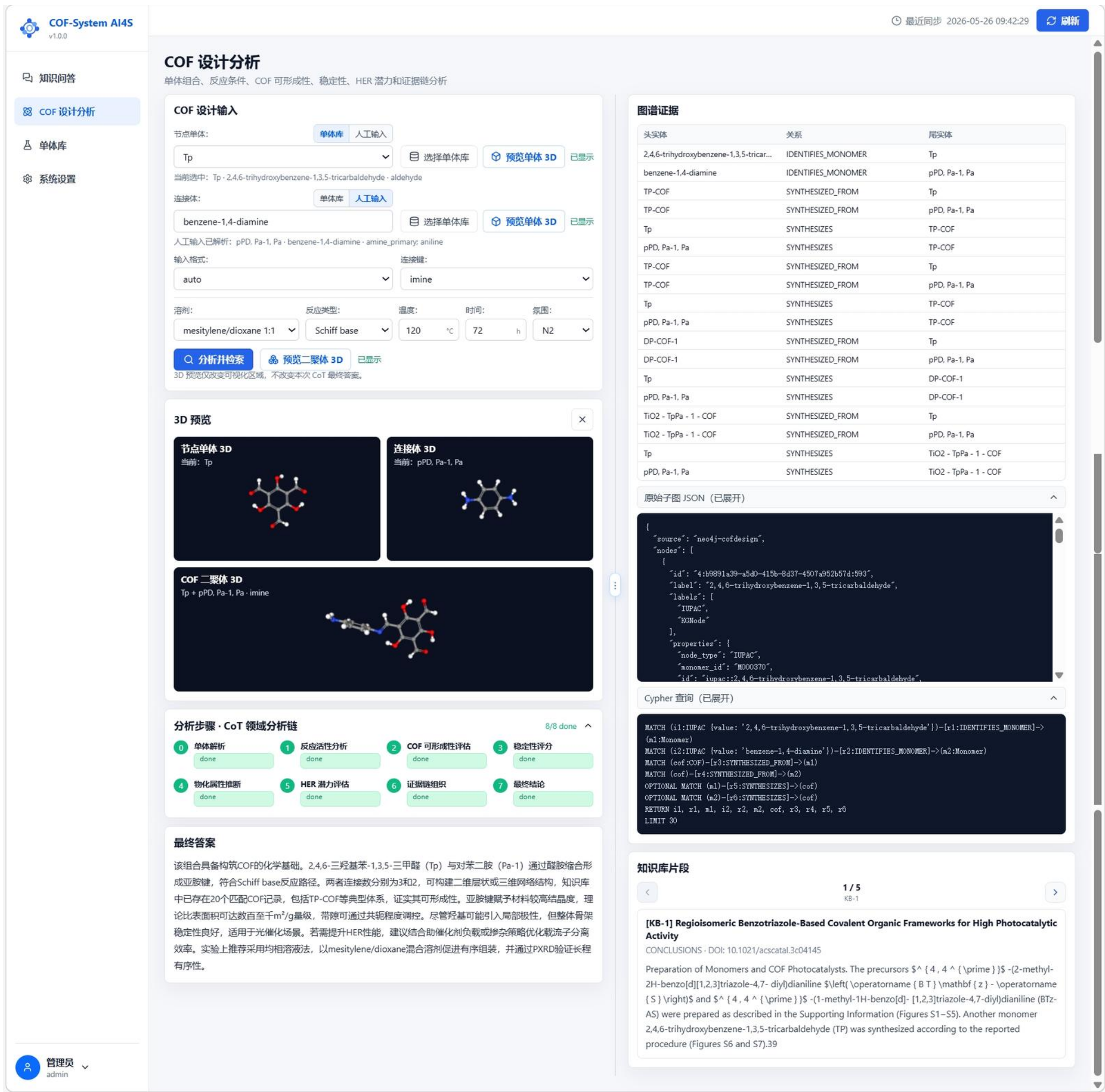
Inspect graph triples, Cypher, raw JSON, and paper fragments.

USE CASE

Use this page when the reviewer needs to see how COF knowledge questions are answered and traced.

2. COF Design Analysis

The design page evaluates whether a monomer combination is chemically plausible for COF construction and surfaces the evidence chain behind the recommendation.



INPUTS

Select monomers, linkage type, and reaction conditions.

3D PREVIEW

Inspect monomer and candidate dimer structures.

ANALYSIS CHAIN

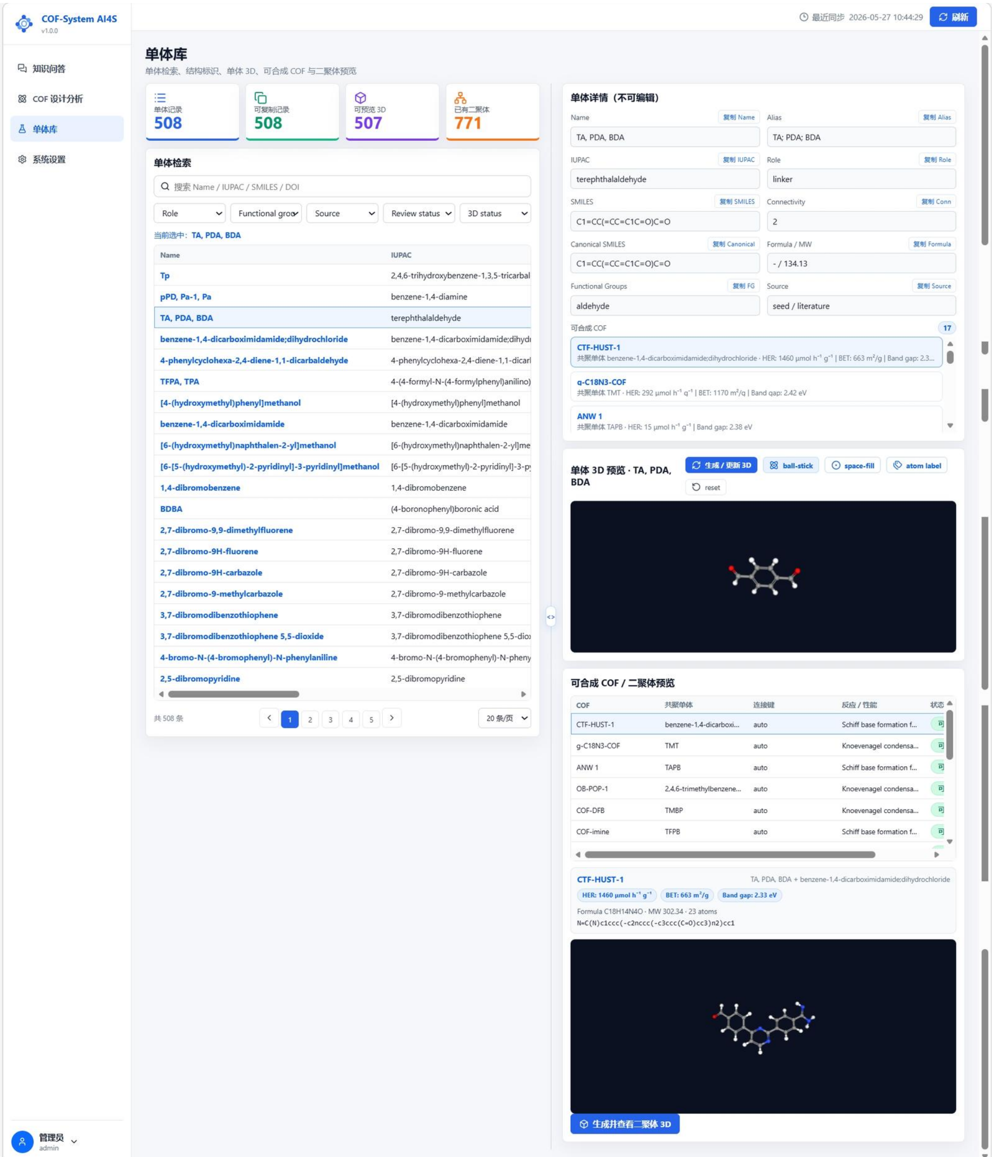
Follow staged checks from parsing to conclusion.

DEMO EXAMPLE

Video example: Tp + pPD/Pa, imine linkage, Schiff-base conditions, 3D preview, and an eight-step evidence-supported analysis.

3. Monomer Library and 3D Preview

The monomer library supplies curated input records for design analysis and lets users inspect structures before using them in a workflow.



SEARCH

Search by keyword and filter by role, group, source, or 3D status.

INSPECT

Review identifiers, metadata, groups, and related records.

PREVIEW

Generate 3D views, switch styles, and inspect dimers.

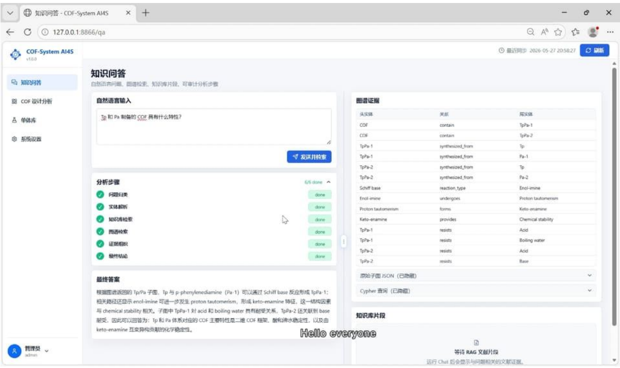
ROLE IN THE WORKFLOW

This interface provides the curated building-block layer for downstream COF design analysis.

Video Walkthrough in Four Frames

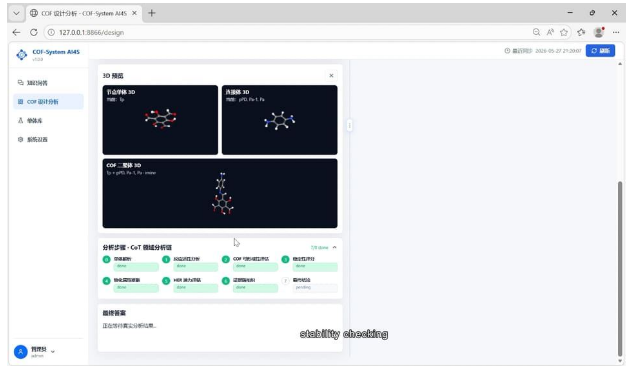
These frames are extracted from the 516-second demonstration video and summarize the reviewer's path through the interface.

01. Ask and retrieve



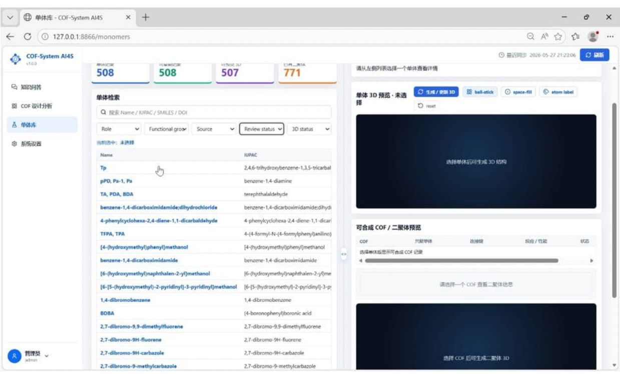
The QA page accepts a domain question and runs the retrieval workflow before displaying a final answer.

02. Preview chemistry



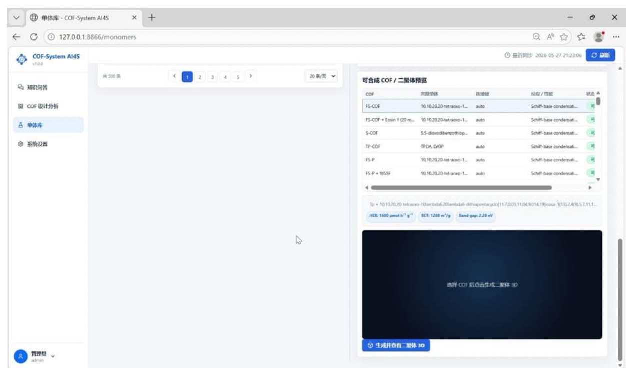
The design page previews selected monomers and a possible dimer before the analysis chain completes.

03. Browse monomers



The library exposes searchable records, functional groups, source fields, and available 3D status.

04. Inspect dimer 3D



Related COF/dimer candidates can be selected and generated for visual inspection.

REVIEWER QUICK GUIDE

The screenshots verify three claims: answers stay paired with graph and literature evidence; design feasibility is presented as a staged analysis chain; and monomer records are connected to 3D molecular previews and related COF/dimer candidates. The HER module should be read as an evidence-aware demo assessment.